

## 2.1: Experiment, Event, Sample Space, and Set theory

## Definition (Experiment (trial))

An experiment is any procedure that can be infinitely repeated and has a well-defined set of outcomes.

## Definition (Sample Space)

The set  $S$  of all possible outcomes of an experiment is called the sample space of that experiment.

## Definition (Countable set)

A set  $X$  is called countable if there exists a one to one and onto function  $f : \mathbb{N} \rightarrow X$ .

Eg1. Show that the set of integers  $\mathbb{Z}$  is countable.

$\mathbb{N}$    1   2   3   4   5   6   . . .

$\mathbb{Z}$    0   1   -1   2   -2   3   . . .

$$f(x) = \begin{cases} \frac{x}{2} & x = 2k \\ -(\frac{x-1}{2}) & x = 2k+1 \end{cases}$$

$f$  is one-to-one and onto hence  $\mathbb{Z}$  is countable.

## Definition (Discrete vs Continuous sample space)

If a sample space  $S$  is finite or countable, we call it a discrete sample space.

If a sample space  $S$  is uncountable, we call it a continuous sample space.

**Eg 2. (Discrete and finite)** Specify the sample space when two coins are tossed.

If we use H for Head and T for Tail. Then

$$S = \{ HH, HT, TH, TT \}$$

Eg 3. (Discrete and infinite) Specify the sample space when you toss a coin until it comes up heads.

$$S = \{ H, TH, TTH, TTTH, \dots \}$$

Eg 4. (Continuous) Consider a Starbucks store. The experiment is to record the wait time of a customer from the time they arrive till they get their order. What is the sample space?

$$S = (0, +\infty)$$

## Definition (Event)

An event  $A$  is a subset of the sample space  $S$ ; we say an event  $A$  has occurred if the outcome is in  $A$ .

**Eg 5.** Consider an experiment in which each of three vehicles taking a particular freeway exit turns left (L) or right (R) at the end of the exit ramp.

(a) Write down the Sample space S.

$$S = \{ RRR, RLL, RLR, RRL, LRR, LLR, LRL, LLL \}$$

(b) What is the event in which exactly one of the three vehicles turns right. Call this event A.

$$A = \{ RLL, LLR, LRL \}$$

(c) What is the corresponding description for this event

$$B = \{ LLL, RLL, LRL, LLR \}.$$

B = At most one vehicle turns right or

B = At least two vehicles turn left

(d) what is the corresponding description for this event

$$C = \{ LLL, RRR \}.$$

C = All the vehicles turn in same direction.

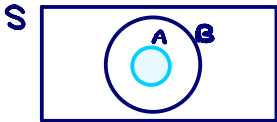
Basic operations in set theory such as intersections, union, or complement is important in probability, since with the help of those we can build new events from existing events. Most concepts from the set theory can be visualized by the use of a Venn diagram. Hence, we will first review some basics from set theory and provide Venn diagram for visualization.

### Definition (Empty Set, $\phi$ )

An empty set is a set that has no element in it and it is denoted by  $\phi$ .

### Definition (Subset)

We say  $A$  is a subset of  $B$ , if every element of  $A$  is an element of  $B$ .  $A$  is a subset of  $B$  is denoted by  $A \subseteq B$ .

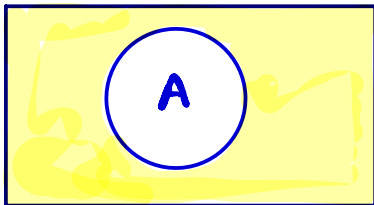


## Definition (Equality)

Two sets  $A$  and  $B$  are equal when  $A \subseteq B$  and  $B \subseteq A$ .

## Definition (Complement)

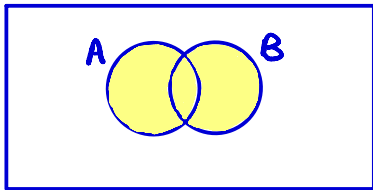
The complement of  $A$ , denoted by  $A'$ , contains every element of sample space that is not in  $A$ .  $A' = \{x \in S | x \notin A\}$ . Some books also denote complement of  $A$  with  $A^c$



The highlighted  
part is  $A^c$

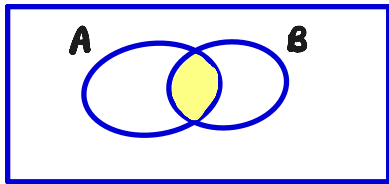
## Definition (Union)

Given sets  $A$  and  $B$  we can make a new set that consists of all the elements of  $A$  and all the elements of  $B$ . This new set is called the union of  $A$  and  $B$  and is represented by the symbol  $A \cup B$ . For example, let  $A = \{a, b, c, d\}$  and let  $B = \{c, d, e\}$ . Then  $A \cup B = \{a, b, c, d, e\}$ .



## Definition (Intersection)

From two given sets  $A$  and  $B$  we can make a new set that consists of all the elements that belong to both  $A$  and  $B$  at the same time. This new set is called the intersection of  $A$  and  $B$  and is represented by the symbol  $A \cap B$ .



## Definition (Mutually Exclusive or disjoint)

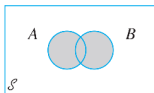
We say two sets  $A$  and  $B$  are mutually exclusive or disjoint if  $A \cap B = \phi$



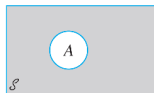
(a) Venn diagram of events  $A$  and  $B$



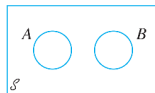
(b) Shaded region is  $A \cap B$



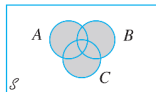
(c) Shaded region is  $A \cup B$



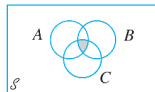
(d) Shaded region is  $A'$



(e) Mutually exclusive events



(f) Shaded region is  $A \cup B \cup C$



(g) Shaded region is  $A \cap B \cap C$

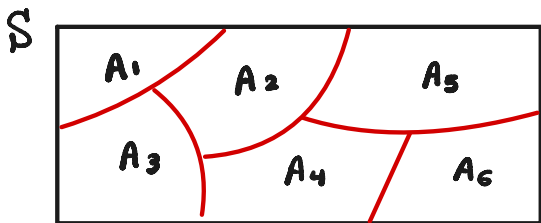
Figure 2.1 Venn diagrams

## Definition (partition)

We say  $A_1, \dots, A_n$  are partition for  $S$  if the following two conditions hold:

(1) they are pairwise disjoint, that means  $A_i \cap A_j = \emptyset$  for every  $i, j \in \{1, \dots, n\}$ .

$$(2) S = \bigcup_{i=1}^n A_i$$



In this example  $A_1, A_2, A_3, A_4, A_5$  and  $A_6$  are a partition for  $S$ .

**Eg 6.** (Problem 6 of Ch 2.1 from Devore) A college library has five copies of a certain text on reserve. Two copies (1 and 2) are first printings, and the other three (3, 4, and 5) are second printings. A student examines these books in random order, stopping only when a second printing has been selected.

(a) List the outcomes in  $S$ .

$$(a) \quad S = \{ 3, 4, 5, 13, 14, 15, 23, 24, 25, \\ 123, 124, 125, 213, 214, 215 \}$$

(b) Let  $A$  denote the event that exactly one book must be examined. What outcomes are in  $A$ ?

$$(b) \quad A = \{ 3, 4, 5 \}$$

- (c) Let  $B$  be the event that book 5 is the one selected. What outcomes are in  $B$ ?

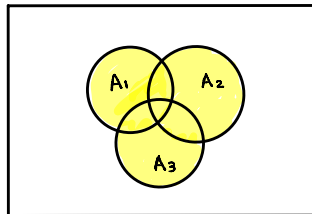
$$(c) \quad B = \{5, 15, 25, 125, 215\}$$

- (d) Let  $C$  be the event that book 1 is not examined. What outcomes are in  $C$ ?

$$(d) \quad \{3, 4, 5, 23, 24, 25\}$$

**Eg 7.** (Problem 8. of ch 2.1 from Devore.) An engineering construction firm is currently working on power plants at three different sites. Let  $A_i$  denote the event that the plant at site  $i$  is completed by the contract date.

(a) At least one plant is completed by the contract date.

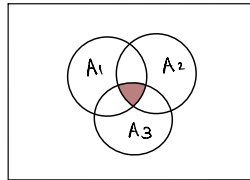


(a) at least one has happend.  $A_1 \cup A_2 \cup A_3$

(b) All plants are completed by the contract date.

all have happened means intersection.

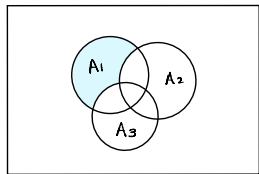
$$A_1 \cap A_2 \cap A_3$$



(c) Only the plant at site 1 is completed by the contract date.

$A_1$  happens and  $A_2$  doesn't happen and  $A_3$  doesn't happen.

$$A_1 \cap A_2' \cap A_3' \underset{\substack{\uparrow \\ \text{or}}}{=} A_1 \cap (A_2 \cup A_3)'$$

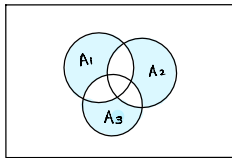


(d) Exactly one plant is completed by the contract date.

$(A_1 \text{ happens and } A_2 \text{ doesn't happen and } A_3 \text{ doesn't happen})$  or

$(A_1 \text{ doesn't happen and } A_2 \text{ happens and } A_3 \text{ doesn't happen})$  or

$(A_1 \text{ doesn't happen and } A_2 \text{ doesn't happen and } A_3 \text{ happens})$



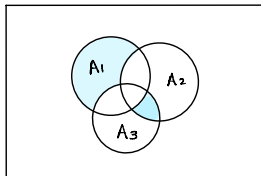
or means union. Hence  $(A_1 \cap A_2' \cap A_3') \cup (A_1' \cap A_2 \cap A_3') \cup (A_1' \cap A_2' \cap A_3)$

(e) Either the plant at site 1 or both of the other two plants are completed by the contract date.

$(A_1 \text{ happens and } A_2 \text{ doesn't happen and } A_3 \text{ doesn't happen})$  or

$(A_1 \text{ doesn't happen and } A_2 \text{ happens and } A_3 \text{ happens})$

$(A_1 \cap A_2' \cap A_3') \cup (A_1' \cap A_2 \cap A_3)$



**Eg 8.** (Problem 4 from Devore) Each of a sample of four home mortgages is classified as fixed rate ( $F$ ) or variable rate ( $V$ ).

(a) What are the 16 outcomes in  $S$  ?

$$a) \quad \left\{ \begin{array}{l} VVVV, FFFF, VFFF, FVFF, FFVF, FFFV, \\ FVVV, VFVV, VVFF, VVVF, \\ VFVF, VVFF, VFFV, FFVV, FVVF, FVFV \end{array} \right\}$$

(b) Which outcomes are in the event that exactly three of the selected mortgages are fixed rate?

$$E = \{VFFF, FVFF, FFVF, FFFV\}$$

(c) Which outcomes are in the event that all four mortgages are of the same type?

$$A = \{VVVV, FFFF\}$$

- (d) Which outcomes are in the event that at most one of the four is a variable-rate mortgage?

$$B = \{FFFF, VFFF, FVFF, FFVF, FFFV\}$$

- (f) What are the union and intersection of the two events in parts (c) and (d)?

$$A \cup B = \{VVVV, FFFF, VFFF, FVFF, FFVF, FFFV\}$$

$$A \cap B = \{FFFF\}$$

- (e) What is the union of the events in parts (b) and (c), and what is the intersection of these two events?

$$E \cup A = \{VVVV, FFFF, VFFF, FVFF, FFVF, FFFV\}$$

$$E \cap A = \emptyset$$

**Table:** Part 1- summary of connection between set theory and probability

| Probability language                    | Notation                         |
|-----------------------------------------|----------------------------------|
| Sample space                            | $S$                              |
| Empty set (impossible event)            | $\phi$                           |
| $s$ is a possible outcome               | $s \in S$                        |
| $A$ is an event                         | $A \subseteq S$                  |
| $A$ occurred                            | (actual outcome) $\in A$         |
| something happens                       | (actual outcome) $\in S$         |
| at least one of $A$ or $B$ occurs       | $A \cup B$                       |
| both $A$ and $B$ occur                  | $A \cap B$                       |
| $A$ occurs, but $B$ does not occur      | $A - B = A \cap B^c$             |
| $A$ or $B$ occurs, but not both of them | $(A^c \cap B) \cup (A \cap B^c)$ |

**Table:** Part2: summary of connection between set theory and probability

| Probability language                                                 | Notation                                                                                    |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| $A$ does not occur                                                   | (actual outcome) $\in A^c$                                                                  |
| at least one of $A_1, \dots, A_n$ occurs                             | (actual outcome) $\in \bigcup_{i=1}^n A_i$                                                  |
| All of $A_1, \dots, A_n$ occur                                       | (actual outcome) $\in \bigcap_{i=1}^n A_i$                                                  |
| If $A$ occurs, then $B$ also occurs                                  | $A \subseteq B$                                                                             |
| $A$ and $B$ are mutually exclusive<br>or<br>$A$ and $B$ are disjoint | $A \cap B = \phi$                                                                           |
| $A_1, \dots, A_n$ are partition for $S$                              | $S = \bigcup_{i=1}^n A_i$ and<br>$A_i \cap A_j = \phi$ for every $i, j \in \{1, \dots, n\}$ |

# Some set-theoretic properties and rules:

$$(A \cup B)' = A' \cap B' \text{ (De Morgan's Law, I)}$$

$$(A \cap B)' = A' \cup B' \text{ (De Morgan's Law, II)}$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \text{ (Distributive Law, I)}$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \text{ (Distributive Law, II)}$$